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Supervised Machine Learning Method for Ontology-based Financial Decisions in the Stock Market

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For changing semantics, ontological and information presentation, as well as computational linguistics for Asian social networks, are one of the most essential platforms for offering enhanced and real-time data mapping, as well as huge data access across diverse big data sources on the web architecture, information extraction mining, statistical modeling and data modeling, database control, and so on. The concept of opinion or sentiment analysis is often used to predict or classify the textual data, sentiment, affect, subjectivity, and other emotional states in online text. Recognizing the message's positive and negative thoughts or opinions by examining the author's goals will aid in a better understanding of the text's content in terms of the stock market. An intelligent ontology and knowledge Asian social network solution can improve the effectiveness of a company's decision making support procedures by deriving important information about users from a wide variety of web sources. However, ontology is concerned primarily with problem-solving knowledge discovery. The utilization of Internet-based modernizations welcomed a significant effect on the Indian stock exchange. News related to the stock market in the most recent decade plays a vital role for the brokers or users. This article focuses on predicting stock market news sentiments based on their polarity and textual information using the concept of ontological knowledge-based Convolution Neural Network (CNN) as a machine learning approach. Optimal features are essential for the sentiment classification model to predict the stock's textual reviews' exact sentiment. Therefore, the swarm-based Artificial Bee Colony (ABC) algorithm is utilized with the Lexicon feature extraction approach using a novel fitness function. The main motivation for combining ABC and CNN is to accelerate model training, which is why the suggested approach is effective in predicting emotions from stock news.

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CCS Concepts: • **Information systems** → **World Wide Web**; **Information retrieval**; **Sentiment analysis**; • **Computing methodologies** → **Artificial intelligence**; **Information extraction**;

Additional Key Words and Phrases: Stock market, opinion mining, sentiment analysis, Lexicon feature extraction, Artificial Bee Colony Algorithm (ABC), Convolution Neural Networks (CNN)

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1 INTRODUCTION

The prediction of the financial market and financial instruments for optimum investment was not an easy task earlier. At the initial level, the prediction of stock markets was made using the efficient market hypothesis and random walk thesis [1]. These theories indicated that it is impossible to predict an accurate stock price trend and other financial instruments [2]. These are based on many crucial factors like domestic and international policy changes, current economic and political scenarios, reports by global agencies (like World Bank, IMF, Moody's), and geo-political developments [3]. It could not be related or derived from their past performances. Therefore, the financial markets will be more randomly driven, and the accuracy of their predictions cannot be more than 50% [4, 5]. This is the same probability of many random events, just like winning a toss in a cricket or soccer match. Later, Baker found that investors' sentiment index affects financial markets [6]. Even the prediction of a candidate's factors for seeking a specific volume of returns is enhanced by the sentiments attached to a particular financial instrument in the market. The key reason behind the success of sentimental analysis is modern-day technology to study these sentiments, investor opinions and feedback, and understanding the emotions and underlying principles of choosing a particular investment instrument [7]. It provides data from the comments and feedback from current investors, experts, and market analysts about any particular company trading in financial markets. Social media's growth has proven to be a perfect catalyst for data availability growth by recording the user's comments, reviews and opinions, Q&As about the market, policy changes, and other necessary developments. Social media, blogs, and rating portals have plenty of information analyzed with intelligent systems by classifying the reviews, opinions, and feedback as negative, positive, and neutral.

Machine learning appears to have caught the interest of academics in recent years. **Convolutional Neural Network (CNN)** is an important deep learning technology for dealing with complicated situations [8, 9]. Current sentimental studies are based on Twitter sentiment analysis for financial markets, as Twitter is a more suited social media platform for discussing financial instruments. Most of the studies have used those datasets that are more specific to their prediction context. This could be one reason for the low accuracy level of predictability. A combinational approach of lexicon-based approaches and support vector machines has been used by authors to achieve a predictability accuracy score of 71% [10]. The primary objective of [11] was to carry out a detailed and comprehensive sentimental analysis of tweets related to the financial market. Researchers have effectively utilized and employed the ensemble method by combining random forest, assist vector machines, degeneration algorithms, and a combinatorial approach to analyze embedded words and lexicons [12, 13]. Using these techniques, the author won first prize in a competition based on sentimental analysis for stock markets. Some regression algorithms are evaluated in this approach; data mining and deep learning methods are used to analyze tweets on financial markets from StockTwits [14]. It was found that their implemented CNN was right, achieving an accuracy rate of 90.8%, while their LSTM accuracy was low at 69.9%.

The stock market dataset in this article is collected from public websites and portals relevant to the stock market, and the repository is known as the Stocktwits Database, which is used for the simulation of the suggested model [15]. From January 2018 to April 2019, 15,000 Twits samples were collected for simulation and validation of the proposed architecture, with 5,000 datasets obtained from each sentiment category, like positive, negative, and neutral [16]. Depending on the Stocktwits Database, sentiment classification is done through the convolution neural networks, and then the opinions thus generated are optimized using the **Artificial Bee Colony Algorithm (ABC)** algorithm to achieve better results. According to the data, the proposed model worked well in terms of categorization. In this scenario, the ABC method is analyzed as a character selection or optimization reach, with the fitness function assisting in training the model totally using CNN as machine learning [17]. Many swarm-based techniques are available for feature selection, but according to the author research in 2016 for the sentiment analysis, ABC is far better than the others. The main contributions of this research are listed as follows:

- To design the lexicon dictionary, which is based on the stock markets' polarity, it twists for feature extraction using ABC as a characteristic selection strategy with a novel aim or fitness purpose to eliminate unnecessary characteristics.
- To train the designed sentiment, categorizing system for the stock market using the concept of CNN as a machine learning approach for negative, positive, and neutral review data.
- Performance measures like precision, recall, F-score, execution time, error, and category perfection are measured and compared with the existing model to validate the proposed system.

1.1 Sentimental Analysis

Sentiment analysis means analyzing the different sentiments/moods of various people. Sentiment can be different for different people; it can be puzzling, diverting, nauseating, or provoking unless one understands what they are about [18]. As sentiments, they are related to feelings and concentrated mainly on opinions and attitudes, and for this reason, sentiments are said to be subjective. Six different categories of emotion are joy, sadness, love, fear, surprise, and anger. Some emotions can be sensed naturally, while some are to be considered for a positive and negative response. Joy and sadness can be sensed using common sense that we deal with daily from different angles [19]. Sentiment analysis helps in providing a positive and negative opinion. This helps remove complexity, since it is too impractical to consider more emotions considering the datasets [20]. The task of recognizing the tone of a text is known as sentiment analysis. A common goal is to classify a text as favorable, negative, or neutral. The method of discovering and analyzing emotions using text input is known as sentiment analysis. Positive, negative, and neutral opinion analysis are the three forms of opinion analysis. Sentiment analysis is a type of feedback that dives deeper into the text to identify material that reflects a customer's feelings about your company. In other words, sentiment analysis finds opinionated perspectives in consumer feedback such as joyful or sad.

SA is categorized into two main classes, i.e., lexicon analysis and machine learning. This is shown in Figure 1. Emotions in the text are detected using this method. It recognizes happy, sad, and furious emotions using lexicons and machine learning. It is a sentiment analysis tool that uses an algorithm to assist a company to figure out why a consumer feels a specific way. When done correctly, sentiment analysis provides a lot of value to a company. Sentiment research provides a deeper understanding of how your consumers feel and what they anticipate from your company. Sentiment analysis is the present and future of all brands, whether new and old. Brands can obtain a good sense of how customers feel about their products and services by employing sentiment analysis. Customer feedback may be analyzed, interpreted, and evaluated using sentiment analysis.

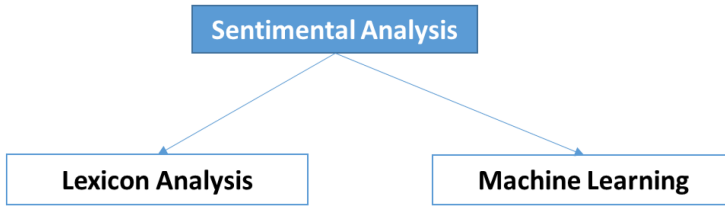


Fig. 1. Sentiment analysis [20].

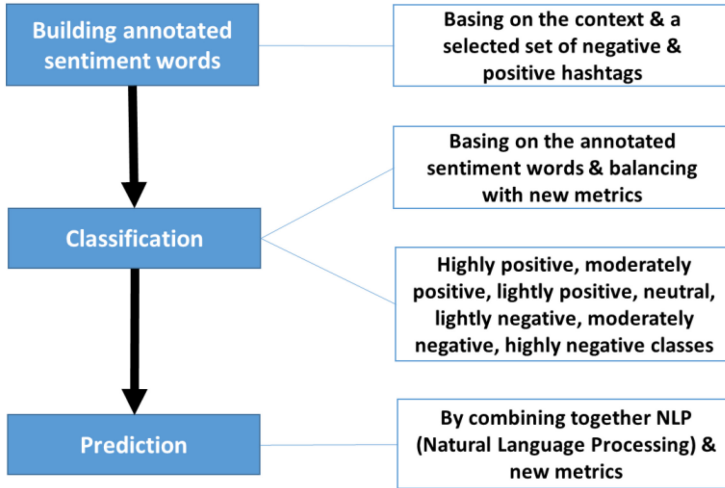


Fig. 2. Stages of the model Carey [21].

Sentiment analysis aids in the discovery of new insights, the comprehension of customers, and the empowerment of staff to be more productive.

Lexicon analysis analysis aims to compute an article's separation from the alignment of words or phrases. The requests created by this type of analysis do not reflect the context analyzed in this section, and machine learning comprises structuring replicas from different datasets to regulate an article's alignment. Studies that use such means can be carried out on a precise topic [21].

The sentimental analysis model consists of three main sets, as described in Figure 2, which builds annotated sentiment words, classification of these sentiment words, and prediction.

1.2 Opinion Mining

The study and analysis of human behavior using a computer modelling technique is known as opinion mining. Opinion mining's key fields include data mining, online extraction, text mining, and so forth. Social media platforms are growing in popularity and are quickly becoming necessary in the lives of the majority of people. Various social networking platforms, such as Facebook, Twitter, and WhatsApp, create a large amount of data, which can be mined to find hidden and important information. Calculating and evaluating average tendencies to any attitude/sentiment to any structure helps both the organization and the individual establish an accurate opinion about current trends or unknown issues. Nowadays, cyberspace is continuously saturated with a plethora of apps and digital media via which people of many backgrounds and experiences may exchange their thoughts and opinions on a wide range of subjects and events. These days, social media sites, like Instagram, Twitter, Telegram, and WhatsApp, and other social media platforms are best for

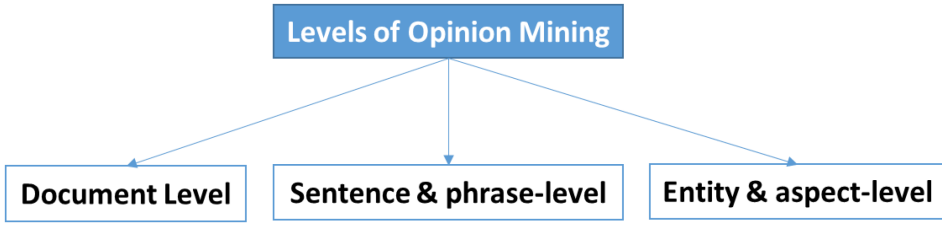


Fig. 3. Process of opinion mining [24].

sharing facts or frequently in text usage. Such shared networking platforms are desirable for exchanging visions or opinions about any product, cinema, or end-user concerning themes by posting comments and pictures and getting feedback from other users. These forms of user initiated blogs, ranging from social networking sites to individuals, serve a crucial role for the government, businesses, and individuals. Information is rigorously gathered from these social networking sites for greater usage.

Text mining is characterized as an augment that uses the following coordinates, i.e., v , x , y , and z where “ v ” is the opinion or sentiment target, “ x ” symbolizes the sentiment related to the goal, “ y ” categorizes the opinion frame, and, last, “ z ” symbolizes the period from where the opinion was specified. Opinion mining is a method to examine and recapitulate the sentiments articulated by vast data. Based on sentiment analysis, opinions can be categorized into three types: positive, negative, and neutral [22]. This process mostly functions in extracting the opinions, categorizing opinions, and analyzing opinions and evaluating them in numerous stages in the form of text. To recognize the sentiment, affect, and other emotional states that are hidden in the text is the main objective of sentiment analysis [23].

Numerous methods are used for to conduct opinion mining. For such a specified area, supervised learning methods are the best to implement, but the model should be able to apply them in another area. Unsupervised learning approaches derive the opinion from service processing language operations by giving a number of criteria for acquiring feature expressions [24].

1.2.1 Level of Opinion Mining. Three levels are used in the opinion mining method. A brief introduction to these levels is explained in this section. Figure 3 presents a different level of opinion mining, categorized into three categories—document level, sentence and phrase level, and entity and aspect level.

- **Document level:** This task’s objective is to learn whether the document states are positive or negative sentiments. The difficulty of document-level sentiment analysis can be observed as a particular case of sorting with natural language data.
- **Sentence and phrase level:** Document-level analysis frequently fragments lesser units that consist of specific words, phrases, or sentences as an intermediate step. In this type of analysis, every term is investigated, and then they are categorized as positive, neutral, or negative. Sometimes, it is complicated to define how a statement of fact should be preserved.
- **Entity and aspect level opinions:** Numerous sentences related to an opinion text might not denote the entity itself. This analysis indicates enormously significant to identify the opinion goals in demand for creating better judgment mining that is strictly valuable.

1.2.2 Discussion on Opinion Lexicon. The number of words represents any opinion word or sentiment word that is usually significant in sentiment analysis and opinion mining [25]. This is

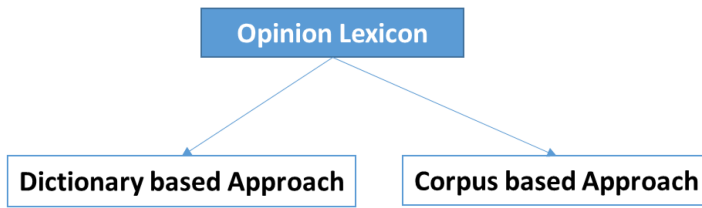


Fig. 4. Opinion lexicon [25].

shown in Figure 4. The opinion lexicon consists of words like *delightful*, *decent*, *immoral*, *tremendous*, and *similar*. Apart from the word, the opinion lexicon can be in the form of phrases like “get out,” “under the table,” “bugger off,” and such. Opinion lexicon can be categorized into two approaches, which can be either be a dictionary-dependent approach or can be a corpus-dependents approach.

(A) Dictionary-based approaches

This method chains together the establishment of words and a dictionary. The primary method is to manually choose an unassertive word that can be either positive or negative. After that, we cultivate the words by checking the vocabulary online via replacement or finding the antonyms of words. This method can be practically changeable for the words that fit into the set and whose replacements or antonyms are to be searched in the dictionary. Additionally, this approach is promising to use, as it helps expand the worth of the words created. Several approaches are being discovered for opinion expansion. The critical drawback of such an approach is that it does not explain the framework of a term for creating judgments. Corpus-based approaches are active in controlling the context [26].

(B) Corpus-based approaches

This approach begins either with positive or negative words. After that, the weight is to be determined by deducing whether other similar words in the neighborhood have positive or negative terms with that. These kinds of techniques, which expand the word set in both positive and negative directions, produce normal behaviour. Adjectives frequently serve both beneficial and detrimental purposes for different parts of a same corpus. Let us consider two words, good or bad, which are different, and hence it is challenging to club them together with “and” operators. We can graph these random words like good and evil related to nodes and links that various subordinate alignments for dealing with difficult situations. Hence as a main contribution this article focuses on predicting stock market news sentiments based on their polarity and textual information using the concept of ontological knowledge-based CNN as a machine learning approach. With the use of the convolution function, the CNN model is employed to create an optimistic forecast. The CNN model is used to identify the variables that will determine whether the stock price is rising or shrinking in the future. The designed technique predicts negative, positive, and neutral sentiment using stock news data taken from the Stock twists database.

Organization: The residue of this article is arranged as follows. Section 2 presents natural language processing and the Asian language network. In Section 3, stock market prediction, related work, Support Vector Machine, and Convolution Neural Networks are presented. Section 4 presents a detailed description of our proposed network and methodology. Thereafter, Section 5 discusses the experimental outcomes and discussion. Eventually, Section 6 concludes the article.

2 NATURAL LANGUAGE PROCESSING

Natural language clarifying attempts to develop machines that can analyze and adapt to text or voice input in the similar way that humans do and then answer in their own text or speech. **Natural language processing (NLP)** is an artificial intelligence area that explores machines' ability to perceive text and speak words of in the similar type as people do. NLP integrates rule-dependent human language modelling in computational linguistics with statistical, machine learning, and deep learning models. When these mechanisms are coupled, computers can analyze human language as text or audio data and "understand" its entire meaning, including the speaker's or writer's intent and sentiment.

NLP is used to empower programmers who can interpret text from one language to the other, reply to audio instructions, and synthesize vast amounts of data in real time. Voice-activated GPS methods, digital assistants, speech-to-text dictation software, customer service, chatbots, and other consumer conveniences all use NLP. NLP is, however, increasingly being employed in corporate results to help firms streamline operations, enhance staff productivity, and simplify mission-critical company operations. Because human language is riddled with ambiguities, designing software that accurately interprets the intended meaning of text or voice input is extremely difficult. Homonyms, sarcasm, homophones, idioms, exceptions to grammar and usage, metaphors, and sentence structure variations are just very few examples of human language ambiguities that take humans years to learn but that programmers must teach natural language-driven applications to recognize and understand correctly from the start if those applications are to be effective.

Many apps have begun to integrate NLP methods to provide a better user experience [27]. Although it is relatively easy to obtain external news from a variety of sources, it is difficult to acquire and analyze data from a company's financial statements. As a result, extracting information substance from financial statements is time-consuming and complicated. Textual information extraction and news story processing date back to 1934 [28, 29]. Characteristics extraction refers to the process of converting raw data into numerical characteristics that may be handled while keeping the information in the original data collection. In terms of results, it outperforms just using machine learning on raw data. Manual feature extraction comprises identifying and summarizing the important attributes in a given circumstance, as well as designing a method for extracting such properties. Knowing the context or domain can assist you in making smart judgments about which attributes are desirable in certain contexts. For decades, engineers and scientists have been working on feature removal algorithms for images, signals, and text. The mean of a signal's window, for example, is a basic property. Automated feature extraction extracts feature from signals or pictures without the need for human interaction, using particular circumstances or deep networks. This method may come in handy when you need to move swiftly from raw data to machine learning algorithms. In computerized feature extraction, wavelet scattering is employed.

2.1 Asian Languages Network

Asia's population is approximately 4.3 billion people. That is almost six times the population of Europe (roughly 739 million). As a result, it should come as no surprise that there is a plethora of additional languages. Furthermore, Asia is home to a wide number of language families. The majority of Europeans are fluent in Indo-European languages, which include Germanic, Romance, and Slavic. Asian languages, however, are classified into various families, which include Indo-European, Sino-Tibetan, Altaic, Dravidian, and others. There is less overlap between today's languages due to the diversity of linguistic families and origins. Finally, Asia's remote regions have allowed for the emergence of various separate languages that are unaffected by outside influences. Asia's way of life is becoming growingly digitized. Social media has become a vital part of the daily lives of millions of people. According to Hootsuite, 64% of the population in Eastern Asia and 55%

Table 1. Various Machine Learning Techniques

Procedure	Category	Material used
Genetic	3	Numerous no of articles from last few years
SVM	2	Approximate 6,982 articles
Naïve Bayesian	4	Approximate 39,562 articles

in Southeast Asia are on prominent Asian social networks. These figures are comparable to those observed throughout Europe. It is a significant benefit for organizations that can use social media listening data to construct a fortunate business plan. However, global social media networks such as Facebook and Twitter are not the only ones popular in this region. The new made-in-Asia media has joined a competitive market, offering far more data regarding Asia Pacific residents. To acquire an accurate view of the statements about them and the industry as a whole, international and Asian enterprises must adapt to these changes in behaviors and follow Asian social media networks in their social listening research.

Stock market Prediction

Forecasting the stock market is the technique of predicting the future price of a stock or other financial instrument [30]. A correct forecast of the future price might end in high returns. The skill to forecast the importance of stock price professionally is vital for market traders or shareholders to make the most of their wages. The Internet is an absurd literal repository of unmanageable statistics, such as business-related broadcasts and Twitter posts. One can customise a large number of literal statistics to acquire data from a variety of industries. Its chief objective lies in calculating the performance of the stock market over several ML techniques and procedures. This model helps resolve many characteristics of the given input and helps predict the stock price, which can be either positive or negative. Several other features may comprise interest rate, car rates, gold rates, diesel and oil rates, foreign exchange rate, live news, a Facebook news feed, and so on.

The stock market attracts educational and specialised studies from all over the world due to its high profit potential; according to present academic models, the stock market is volatile [31]. Only the most obvious of these educational concept statements, the efficient market hypothesis, is intolerable to tyre out the market. Regardless of educational theories, researchers are still confident in forecasting the stock market due to the motivation of unbounded money. With the assurance of limitless wealth, plenty of research has been done to forecast the stock market even though none of these studies have been wholly positive, showing that the stock market is not as significant as formerly assumed.

2.2 Related Work

Machine learning [32] delivers approaches for finding designs in large data volumes. Most of the research has been inspecting the numerous procedures that can lead to stock price prediction. Table 1 explains the resulting prediction of stock market classification using dissimilar machine learning mechanisms.

Machine learning is a leading technology in computer science that provides machines with the ability to acquire some knowledge. There are two categories of ML algorithms, i.e., supervised learning and unsupervised learning.

(A) Supervised Learning

In this type of learning, various inputs and related labels are taken to train the machine learning algorithm. The models try to be close to $y = f(x)$ to get better results. The input is denoted by x , and y denotes labels. For supervised learning, it is mandatory to train the dataset with accurate labels. In this type of learning, various algorithms are constructed on

the basics of production variables. Production will be called a *regression task* if the variable is constant in the algorithm. An example of a regression task may include the estimation of house prices, and so on. Production will be called a *classification task* if the variable keeps on changing in the algorithm. Examples for this may include type, color, shape, and so on. Supervised learning methods are utilized in a wide range of machine learning applications. Linear regression, random forests, support vector machines, and logistic regression are some instances of supervised learning.

(B) Unsupervised Learning

In this type of learning, the input is provided, but no output labels will be provided. In this learning approach, the algorithm tries to configure the inputs provided to it and notice the output design. This learning approach can be characterized in two tasks, mainly cluster and association. In the association rule mining processes like precedence attempts to forecast consumers' upcoming consumption manners while in clustering, various processes attempt to learn integral clusters or groups in the information.

In [33] employs a Twitter mining technique to classify people and their textual content based on domains. The proposed technique includes machine learning modules. Experiments were carried out to validate the mechanism described in the research framework, which was backed up by the great performance of the evaluated metrics. The trial validated the framework's applicability to effective domain-dependent categorization for Twitter users and their material, as evidenced by the findings of various performance assessment metrics. The work suggested a concept-based, ontology-based automated classification system with minimal concepts. These ideas, which are essentially the low-level concepts of an ontology and the instances of the ontology obtained via ontology reasoning, represent the text [34]. Background information produced from ontologies is widely used to enrich papers with semantics from the biomedical domain [35]. Researchers proposed an **interactive artificial bee colony (IABC)**. The Newtonian rule of universal gravity was included into the onlooker phase of IABC, in which onlookers are selected using a roulette wheel [36]. Ontology-dependent document categorization has become an increasingly relevant and intriguing study topic in a variety of domains, particularly in the enrichment of documents and classified representations provided via the usage of ontologies. The study shows an example of document categorization that employs ontologies and document relationships as background information to augment the document representation [37]. Several academics have suggested using evolutionary computing methods to automatically construct the best CNN designs and improve their performance [38]. Reference [39] employed an evolutionary technique to automatically build CNN designs that outperform state-of-the-art architectures. Authors have investigated different evolutionary computing-based methodologies for optimizing CNN designs, which were tested on benchmark datasets and produced a competitive outcome [39]. The study has developed an approach for autonomously constructing CNN structures using genetic algorithms in conjunction with grammatical evolution. To prevent and combat fraud threats, we suggest combining machine learning with an ontology based on an electronic payment system and another ontology developed automatically [40]. Through tracking and simulation, this study provides a feasible framework with substantial implications for a range of business-related applications, including the voices of the client of the marketplace, recommendation systems, the discovery of domain-based influencers, and opinion mining [41].

2.3 Support Vector Machine

One of the classes of supervised learning is **support vector machine (SVM)**, which depends on measurements. For jobs like regression and classification, this class of supervised learning can be used. In SVM, each dataset belongs to n -dimensional space, which relates the story of dimension.

The story helps in describing the position on the consistent axis. Two operations are performed; during training time, SVM divides the input layer, and during testing time, the groups of unlabeled data are taken up for plotting and inspecting each side of the hyperplane.

Recent research definitely indicates that the SVM assist is the best stock trend forecasting approach. It is a classification method. Classification procedures detect datasets, create the modifications discovered among classes, and decide which class the incoming data points will be assigned to. SVMs are known to handle vast amounts of clutter data better than other machine learning methods. A substantial amount of prior study has been conducted to evaluate several machine learning algorithms on a particular dataset to regulate the statistically substantially enhanced technique for explaining this specific problem. With the old rates of Google and Yahoo, several researchers discovered the estimating ability of four different algorithms. This training surveys seven daily variables over two years: Open price, Close price, Trade Volume, Date, High price, adjusted close price, and low prices [43].

2.4 Convolution Neural Network

At present machine learning is used in every corner of technology, but recently deep learning has attracted attention, mainly in the field of imaging and ordering. Deep learning is a type of machine learning that employs a three- or more layered neural network. These neural networks seek, albeit with little success, to simulate human brain function by allowing it to “learn” from large volumes of data [44]. Deep learning is based on machine learning. Deep learning is divided into two parts: convolutional neural networks and recurrent neural networks. The input for convolutional neural networks will be a two-dimensional (2D) image, which does not imply that the model may be altered to detect future results. We can identify the results for the future in two ways. One method is to convert the 1D input data into a 2D matrix, while another is to provide the benefits of 1D for result calculation. CNN is one of the feed-forward neural networks. Like earlier neural network models, which may consist of input layer, hidden layer, and output layer, CNN comprises various features like the earlier existing model. The output generated in the CNN layer becomes an input to the other ways corresponding to other layers. The convolutional neural network may consist of numerous other benefits like full connection layers, pooling of layers, and so on. A hidden layer in the convolutional neural network makes it different from others in an existing neural network. The higher the number of hidden layers, the better the results compared to other existing models. These days, there is much work done in the field of CNN. As the amount of textual data grows at an exponential rate, developing models to interpret text data automatically becomes increasingly vital. Gender, age, nationality, mood, and other descriptors may be present in the writings. Many studies of text categorization have developed as a result of the use of such labels in various industrial domains. The CNN model has recently been adopted for the job of text categorization, with promising results. As the number of textual data grows at an exponential rate, developing models to interpret text data automatically becomes increasingly vital.

2.5 Stock Prediction Based on Convolution Neural Network

With the help of the convolution function, the CNN model is used for building an optimistic prediction. The CNN model is used to achieve the values that will tell the stock’s prediction and whether it will increase or decrease in the future. For controlling the model, various teaching numbers will be taken into account. A vector of stocks will be taken as the input for this model to establish a stock prediction model. In computing the results for the prediction of the stock, the conventional function will be considered. The model used for stock prediction is linear, so attempting to solve any nonlinear problem will be very difficult. For solving such problems, the activation function is required. For the gradient vanishing related problem, the activation function should be identified,

and a proper function should be chosen to help find the activation function results. The function used for this model is listed below [45],

$$Relu(x) = \begin{cases} 0 & (x \leq 0) \\ x & (x > 0) \end{cases}, \quad (1)$$

$$Lrelu(x_i) = \begin{cases} a_i x_i & (x \leq 0) \\ x_i & (x > 0) \end{cases}. \quad (2)$$

In this function, x and x_i are the results of the computation of the activation function and a_i is the hyperparameter. Initially, the value of a_i will be taken to be 0.01. Feature mapping is performed after the activation function has been computed. Various pooling layers will be used as input to the convolutional layer. The input will aid in the acceleration of results, complications, and the reduction of feature dimension. The pooling function will be explored for such work to produce better outcomes. The convolutional layer and pooling layer will be computed next. These outputs will become the input for the completely connected layers. This fully connected layer can help in various tasks of classification. Last, the prediction of the stock will be computed, and the computation of the average error will also be recorded. There should be a minimum error as possible; this is the only thing that should be considered. Accuracy of the model will depend upon the number of errors that we have figured out. Two essential steps should be considered for achieving better accuracy results. The first step is to compute the fully connected layer results, which can be computed using formal ones,

$$y_i = \frac{e^{a_i}}{\sum_{k=1}^C e^{a_k}}, \quad i = 2, \quad (3)$$

where

i is the quantity

a is output generated from the fully connected layer

c is the quantity of a and e .

The second step is to calculate the error function using the following formula:

$$J(W) = - \sum_{k=1}^n \sum_{i=1}^C t_{ki} \log(y_{ki}), \quad (4)$$

where

n is sample quantity

C is output of the fully connected layer

t_{ki} is chance through which k belongs to class i

W is the weight coefficient of the filter.

3 PROPOSED SCHEME

The inferences drawn from the existing literature are listed as follows:

- (1) The data used by the researchers in the literature are standard, and no work is done on user-defined data.
- (2) Existing preprocessing cannot yield normalized data, and the possibility of an irrelevant feature set is increased due to the advent of un-normalized data, data with punctuation, and stop words.
- (3) Several optimization algorithms have been used to remove the above problem; however, the researchers did not achieve great accuracy in optimizing the text data.

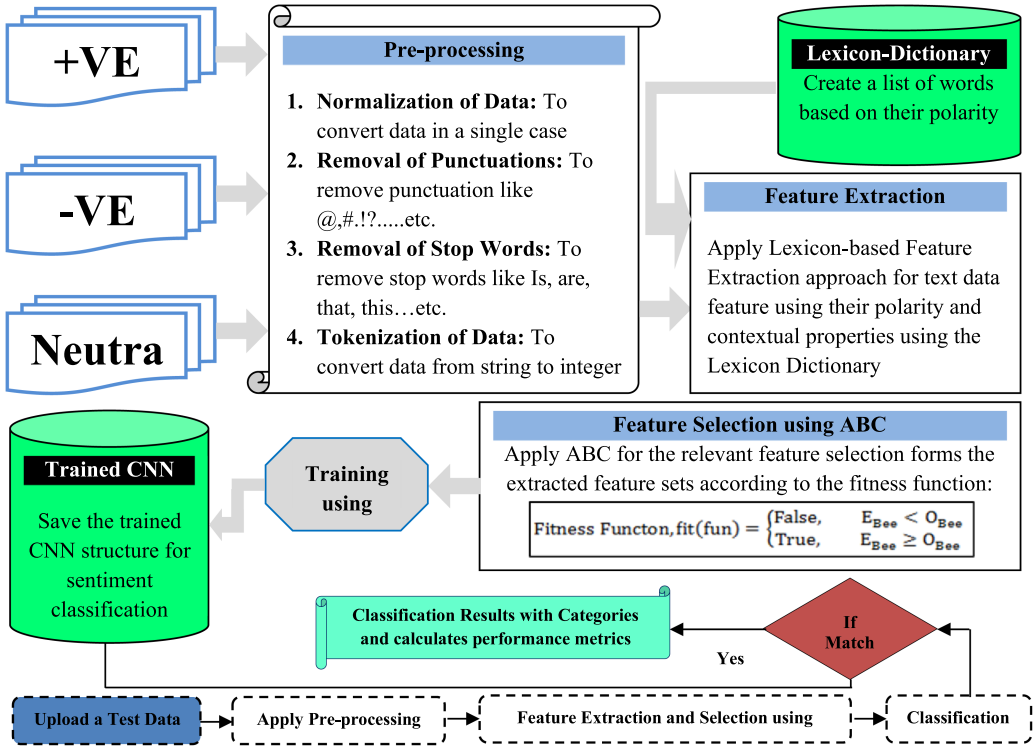


Fig. 5. Architecture of proposed model.

- (4) The majority of current research is based on binary classifiers such as Naive, Bayes, Support Vector Machine, K-NN, and others. The system's complexity is higher for multitype sentiment/opinion classification, and it cannot perform better on the large database.
- (5) Individual scholars have employed the concept of clustering with optimization approaches in the literature; however, clustering algorithms are unsupervised and act on the estimated centroid. If the centroid value varies, then the likelihood of irrelevant findings increases.

3.1 Dataset

There are various online repositories for data analysis, and predictive modeling is available, and to simulate the designed model, we consider Stocktwits Database for the simulation of the proposed model. Here 15,000 Stocktwits samples are taken from January 2018 to April 2019, and 5,000 datasets are taken from each category, such as positive, negative, and neutral. In this article, the stock market dataset is collected from public websites and portals related to stock market, and then rigorous preprocessing is done on the dataset. Sentiment classification is done through the convolution neural networks, and then, finally, the opinions thus generated are optimized using the ABC algorithm to achieve better results. The previous year's stock data are taken to train the model, and specific machine learning libraries are used to clean the data to analyze and predict it. The dataset is collected from different public repositories available on the internet; 80% is used for training, and 20% is used for testing the model. MatLab is used for data processing, which then prepares the data for feature extraction. As a feature, the date and closing price of a stock on a particular day is selected. The same is trained in this model. The proposed model makes use of an optimization algorithm to achieve good classification performance. Initially, the dataset was pro-

cessed in the raw form through feature extraction, followed by a classification process for splitting into different segments.

3.2 Proposed Methodology

In lots of existing work, the designed system only consists of sentiment analysis that aims to use automated tools to detect personal information such as opinions, attitudes, and feelings regarding positive and negative polarities expressed in the text. The key source of concern is that the sentiment extraction procedure is not generic but rather domain specific. A person's vocabulary, language, and resources must be a domain appropriate to provide meaningful results. There is a problem with subjectivity and objective text. It is extremely difficult to quantify the polarities, intensifiers, negatrons, and computed sentiment of rating words. The customized polarity scale, as well as the mathematical discoveries derived from computations, must be translated into something relevant and concrete for the end-user. In the proposed work, analyzing the sentiments of stock market review system using lexicon-based feature extractor and ABC and CNN is designed. Here, CNN's concept is used as a classifier to train the proposed system that helps in the sentiment analysis classification phase. According to the class, the system's training depends on the relevancy of the feature sets, and appropriate feature sets would train CNN. Therefore, in the classification section, stock market data can be easily classified with a trained structure regarding positive, negative, and neutral polarity. The architecture of the proposed system is shown in Figure 5. The stepwise execution of the proposed model is discussed as follows:

- Step 1:** Design a framework for sentiment analysis/opinion mining and upload data with positive, negative, and neutral for the training and testing of the proposed system.
- Step 2:** In both sections, we preprocess the submitted data. Pre-processing involves a number of operations to generate data that fulfils the standards. Text preprocessing may improve the quality of feature extraction as well as the outcomes of sentiment analysis/opinion mining.
- Step 3:** Both the training and assessment phases will use preprocessing. The feature extraction technique is then applied to the preprocessed data to extract feature sets from positive, negative, and neutral data using a lexicon-based lexicon. A lexicon-based technique is employed on the tokenized data after preprocessing, and the algorithm used for lexicon dictionary-based feature extraction is written as:
- Step 4:** Feature optimizations are a practical approach to better accuracy in sentiment analysis/opinion mining. It is used to remove the unwanted feature sets using the novel objective function. As a feature optimization technique, ABC is used in the extracted lexicon-based feature sets. The algorithm of the ABC for feature selection is given as:
- Step 5:** After the feature optimization, the CNN classifier is initialized to train the system based on the optimized data using the following steps:
 - Select an optimized feature as an input of CNN for training and testing.
 - Compute the total emotion/opinion categories (positive, negative, and neutral) generated by optimized data using classifiers
- Step 6:** Then, in the classification stage, categorize the test data using the training structure of the classifiers.
- Step 7:** To validate the proposed system, the performance parameters of the proposed sentiment analysis/opinion mining system such as precision, recall, F-measure, Execution Time, and accuracy are determined at the end of the module. Figure 6 depicts the flow of the proposed work. The suggested model consists of two stages. The first is to upload the data in the database for training purposes and upload the test data. Initially, various steps have been carried out like preprocessing, feature extraction, and feature optimization with some activation functions for uploading the database for training. After that, the datasets are

ALGORITHM 1: Lexicon-based Feature Extraction**Input:** STD→Pre-processed of Stock Text Data**Output:** TFS→Text Feature set of STD**Start feature extraction**

Load, LFE Dataset = Lexicon-based Feature Extraction Dataset

Calculate size of Pre-processed Stock Text Data (R, C) = Size (STD)

For i→1 **to** R**For** j→1 **to** C**If** STD (i, j) = **Positive Polarity** according to LFE Dataset

Positive Feature (i, j) = data (i, j)

Else if data (i, j) = **Negative Polarity** according to LFE Dataset

Negative Features (i, j) = data (i, j)

Else

Neutral Features (i, j) = data (i, j)

End – If**End – For****End – For**

TFS = [Positive Features Neutral Negative Features Features]

Return: TFS as the Feature set of STD**End – Algorithm****ALGORITHM 2:** ABC as a Feature Selection**Input:** TFS → Extracted Feature Set of Stock Text Data

FF → Designed fitness function for feature selection

Output: OFS → Optimized Feature Set**Start selection of features****To optimized the TFS, ABC is used****Set up the basic parameters of ABC:** Bee population (B_p) = Number of TFS DataDefine Fitness function for the selection of the best B_p,

$$\text{Fitness Function: } F(f) = \begin{cases} \text{False; if } E_{Bee} < O_{Bee} \\ \text{True; if } E_{Bee} \geq O_{Bee} \end{cases}$$

In the fitness function, E_{Bee} is pre-processed current text data featured as an employee bee and the O_{Bee} is on looker bee, and it is the average of all B_p

Calculate the length of TFS in terms of Row and Column, [Roz, Column] = Size (TFS)

Set OFS = [] // Initiate an empty variable to store selected data**For** i = 1→ Row**For** j = 1→ Columnb_{ee} = B_p (i, j) // Current bee from B_p $O_{Bee} = \text{Threshold}_{B_p} = \sum_{i=1}^{\text{Row}} \sum_{j=1}^{\text{Column}} B_p(i, j)$ // Mean of all B_p $F(f) = \text{Fit Fun}(E_{Bee}, O_{Bee})$

OFS = ABC (F (f), TFS (i, j))

End – For**End – For**

Check the index of OFS

If OFS (index) = True

OFS = Select (TFS)

Else

OFS = Reject (TFS)

End – If**Returns:** OFS as an Optimized Feature Set**End – Algorithm**

ALGORITHM 3: Train-Net CNN

Input: OFS $\rightarrow$ Optimized Feature Set of text data as trainingdata
Class $\rightarrow$ Review class in Stocktwist dataset (Categories like +VE, -NE, and Neutral Stock market news reviews)
Output: Train-Net $\rightarrow$ Trained structure with Classified PolarityResults
Start
Initialize the Pattern net-based CNN with OFS:
Number of Epochs (E) // Iterations used by CNN
Number of Neurons (N) // Used as a carrier
Performance: MSE, Gradient, Error Histogram and Validation
Data Division: Random
Calculate Length of FS, Len = Size (OFS)
For i = 1 $\rightarrow$ Len
If OFS belongs to +VE
Group (1) = Feature of OFS +VE Class Review
Else if OFS belongs to -VE
Group (2) = Feature of -VE Class Review
Else
Group (3) = Feature of Neutral Class Review
End - If
End - For
Initialized CNN using OFS and Group
Train-Net = CNN (N)
Set the training parameters according to the requirements and train the system
Train-Net = Train (Train-Net, OFS, Group)
Test Review Data = Sim (Train-Net, Current Stock Review Feature Sets)
If Test Data = 1st Class
Classified Results = +VE Review
Else if Test Data = 2nd Class
Classified Results = -VE Review
Else
Classified Results = Neutral Review
End - If
Return: Train-Net as a trained structure with Classified Polarity Results
End

trained using CNN and finally stored in a database that will be used later when we need to extract the information regarding that element. If the element gets matched, then the classification of the result into categories is to be done, and the calculation of the performance parameters is done. However, if not matched, then only calculation of the performance parameter will be done, and the process will stop.

4 RESULTS

This section consists of various screenshots for our proposed model, and the comparative analysis between various parameters has also been briefed here.

4.1 User Interface

Figure 7 represents the screenshot of the output window. The output window has two panels, the training panel and the testing panel. In the training panel, there is a training button; clicking it trains the dataset. However, in the testing panel, the upload test data button helps upload the

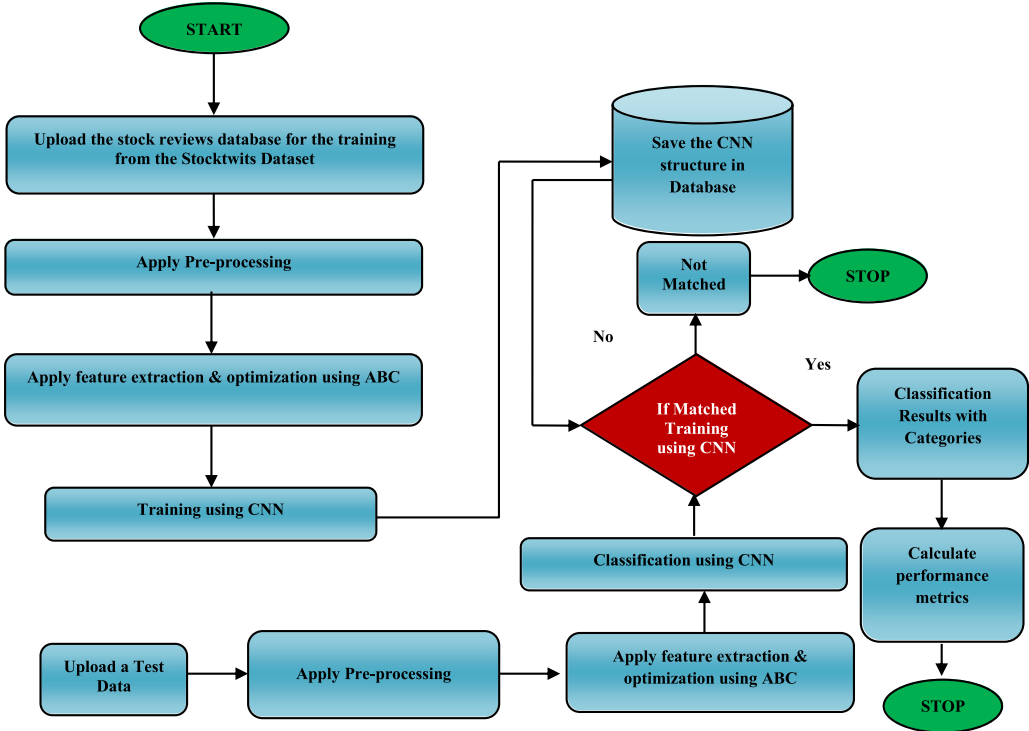


Fig. 6. Flow of proposed model.

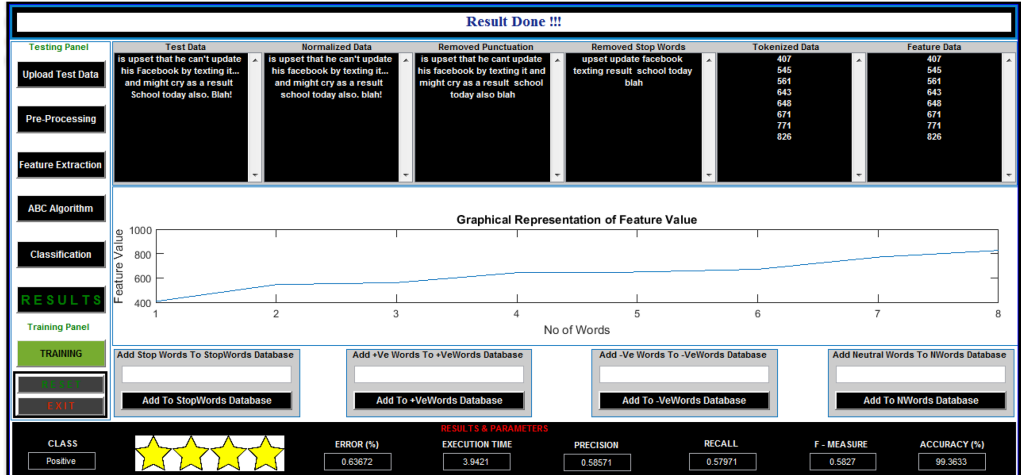


Fig. 7. User interface of the proposed model.

dataset, which can be seen in the test data box. By clicking the preprocessing button, the test data are normalized. All uppercase letters are converted to lowercase letters, which are seen from the normalized data box. A removed punctuation box shows the uploaded test dataset after removing the full stop, exclamatory sign, and so on.

Table 2. Average Results for Different Parameters

Parameter	Value
Error Percentage (%)	0.636
Execution Time (secs)	0.44
Precision	94.62
Recall	92.83
F-Measure	92.00
Accuracy (%)	99.85

Table 3. Comparison Based on Error Percentage

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	0.63672	0.3374	0.57524
2	1.3085	1.5737	0.70339
3	0.017987	1.6543	0.067773
4	0.87	1.4897	0.41545
5	0.080818	0.62389	1.833

It was followed by the removed stop words box that removes all stop words present in the uploaded dataset, e.g., that *his*, *he*, *by*, and so on, and thereby generates the tokenized data, i.e., assigning the token values.

The feature extraction button then represents the feature data values seen from the last text box labeled feature data. When the ABC button and classification button are clicked, it works on the code window, and when the processing is executing, the message is displayed in the output window. The results button, when clicked, includes a class of the dataset, i.e., whether the dataset is positive, negative, or neutral, along with the values of the parameters like error percentage, execution time, precision, recall, F-measure, and accuracy, as shown in Table 2.

In Table 2, the proposed model shows the resultant values of a set of parameters taken into consideration. The proposed model has achieved a minimum error of 0.636 with an execution time of 0.44 seconds and a maximum precision value of 94.57. It has achieved a recall value of 93.72 with an F-measure value of 92.85, along with 99.98% accuracy.

4.2 Evaluation Parameters

A comparison among positive, negative, and neutral datasets is made considering the parameters mentioned afore, which are listed below.

4.2.1 Error Percentage. Error percentage is the difference between a measured and known value, further divided by the known value and multiplied by 100%,

$$\text{Error percentage} = \frac{\text{Recognized value} - \text{investigational value}}{\text{recognized value}} \times 100\%. \quad (5)$$

A comparison of parameter error percentage of different datasets (positive, negative, and neutral datasets) is shown in Table 3. Five instances have been taken for each dataset and, based on these instances, comparisons have been made successfully.

Table 4. Comparison Based on Execution Time (Seconds)

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	0.608	0.53137	0.39902
2	0.66058	0.46148	0.36172
3	0.48012	0.38534	0.48002
4	0.44482	0.37639	0.37615
5	0.75441	0.375	0.36342

Table 5. Comparison on the Basis of Precision

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	94.57	94.29	93.57
2	92.29	91.29	94.86
3	94.14	93.14	92.29
4	94.57	94.43	91.00
5	93.71	93.57	94.57

The minimum error value 0.017 is achieved using the proposed model on the positive dataset. The minimum execution time value on the negative dataset is 0.33 and on the neutral dataset is 0.06.

4.2.2 Execution Time. Execution time is the one during which a program is running compared to other program lifecycle phases such as compile-time, link-time, and so on.

Table 4 compares the positive, negative, and neutral datasets based on the execution time (in seconds) parameters. The minimum execution time value of 0.44 seconds is achieved using the proposed model on the positive dataset. The minimum execution time value on a malicious dataset is 0.37 seconds and on the neutral dataset is 0.36 seconds.

4.2.3 Precision. Precision is the portion of related instances among the retrieved instances.

$$\text{Precision} = \text{true positive value} / (\text{true positive value} + \text{false positive value}). \quad (6)$$

Table 5 presents the comparison of positive, negative, and neutral datasets based on the precision parameter. The highest precision value, 65.71, is achieved using the proposed model on the positive dataset. The highest precision value on the malicious dataset is 67.14 and on the neutral dataset is 64.28.

4.2.4 Recall. The recall is the proportion of relevant instances retrieved compared to the total number of relevant instances. Precision and recall are thus reliant on an awareness of and measurement of relevance.

$$\text{Recall} = \text{true positive value} / (\text{true positive value} + \text{false negative value}). \quad (7)$$

Table 6 presents the comparison of positive, negative, and neutral datasets based on the recall parameter. The highest recall value, 66.66, is achieved using the proposed model on the positive dataset. The highest F-measure value on the malicious dataset is 64.61 and on the neutral dataset is 65.2.

4.2.5 F-Measure. The F-measure parameter measuring a test's correctness is the average of precision and recall. It considers the precision and recall values for calculating the final score,

$$F = (2 \times \text{precision} \times \text{recall}) / (\text{precision} + \text{recall}). \quad (8)$$

Table 6. Comparison on the Basis of Recall

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	90.97	90.77	90.33
2	88.93	90.29	90.77
3	92.33	92.62	91.28
4	93.72	91.97	92.16
5	90.67	90.97	90.97

Table 7. Comparison Based on F-Measure

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	89.88	88.03	89.92
2	90.02	90.29	92.31
3	90.85	90.86	87.78
4	92.10	89.70	90.54
5	92.85	91.27	90.27

Table 8. Comparison Based on Accuracy

S. No	Positive Dataset	Negative Dataset	Neutral Dataset
1	99.36	99.66	99.42
2	98.69	98.43	99.30
3	99.98	98.35	99.93
4	99.13	98.51	99.58
5	99.92	99.38	98.17

Table 7 represents the comparison of positive, negative, and neutral datasets based on the F-measure parameter. The highest F-measure value, 66.18, is achieved using the proposed model on the positive dataset. The highest F-measure value on the malicious dataset is 65.85 and on the neutral dataset is 64.77.

4.2.6 Accuracy. The accuracy can be called the degree to which the outcome of a capacity, design, or requirement conforms to the correct value or a standard. In other words, accuracy describes the boundaries of the errors that can be made when the instrument is utilised under typical working conditions.

$$\text{Accuracy} = \frac{\text{True positive} + \text{true negative}}{\text{True positive} + \text{true negative} + \text{false positive} + \text{false negative}} \quad (9)$$

Table 8 represents the comparison of positive, negative, and neutral datasets based on the accuracy parameter. The highest accuracy, 99.98, is achieved using the proposed model on the positive dataset. The highest accuracy on the malicious dataset is 99.66 and on the neutral dataset is 99.93.

A confusion matrix, also known as an error matrix, is a table that is used to assess the performance of an algorithm. It is usually utilized in supervised learning, although it is also known as the matching matrix in unsupervised learning. Two parameters are used, namely actual and predicted, as shown in Table 9. For the measurement of the accuracy of the model, in the present study, a total 100% is used to measure the accuracy of the method. In the first case, the prediction is 99.36% and the actual is also 99.36%, which is called the true positive. In the second case, the prediction is

Table 9. Confusion Matrix

Actual	Predicted				
	Total 100	Yes	No	Yes	No
		Yes	No	Yes	No
	Yes	99.36 (TP)	98.43	98.69 (TP)	98.51
	No	99.13 (TN)	98.36	98.35 (TN)	99.38

Where TP is the True Positive and TN is the True Negative.

Table 10. Comparison with Existing Techniques

Model	Accuracy	Precision	Recall	F-measure
[12]	90.93	91.68	90.04	90.86
[37]	87.62	87.60	87.45	87.50
[46]	76.68	76.8	75.3	-
[13]	94.4	84.7	74.5	79.3
[52]	79.08	79.23	79.01	79.12
Proposed model	99.85	94.62	92.83	92.00

99.36% but the actual is also 99.13%, which is called true negative; similarly, prediction and actual are given for other cases.

4.3 Comparison with Existing Works

In our proposed study, a classification technique called CNN is employed, along with an optimized algorithm, to train a vast amount of dataset that is optimize during an artificial bee colony algorithm. When compared to previous studies done by other researchers, this technique can attain higher accuracy [47]. The artificial bee colony method was put forward by the author and has been used to tackle numerical optimization issues. This algorithm was inspired by the behavior of honey bees when hunting for a good food source. The ABC approach is a simple, fast, and easy-to-use population-based stochastic search strategy [48]. In the ABC algorithm, the artificial bee colony is composed of three categories of bees: employed bees, bystanders, and scouts. Half of the colony is made up of fake bees, while the other half is composed of spectators. Each food source only has one bee. The employed bee gets turned into a scout after the other bees depart the food source. The nectar amount of a food source correlates with the quality (fitness) of the related solution in the ABC technique, and the placement of a food source indicates a feasible solution to the optimization problem [49].

In comparison with various population-based algorithms, the ABC algorithm has substantial advantages. The first advantage is that it is simple to use, adaptable, and timely. Second, it employs fewer control parameters than other methods such as Particle Swarm Optimization and the Genetic Algorithm [50]. The third advantage of the ABC Algorithm is that it may be easily integrated with other optimization approaches. The fourth advantage of the ABC Algorithm is its great robustness and speed of convergence. The ABC technique is also capable of dealing with stochastic cost objectives. The ABC approach, however, frequently encounters premature convergence in the latter search period. Occasionally, the best deal cannot be found.

Sentimental analysis of the financial mechanisms governing the stock market is carried out in [51] using convolutional neural networks, deep learning, and opinion mining. The proposed model performs better than current state of art methods. In many parameters, proposed model has demonstrated improved outcomes. The comparison between the current work and the proposed work is shown in Table 10.

Table 11. Comparison with Existing Models in Terms of Accuracy

Model	Accuracy
[50]	92.70
[17]	90.89
[34]	59.15
[51]	60.33
[52]	89.50
[53]	91.67
Proposed model	99.36

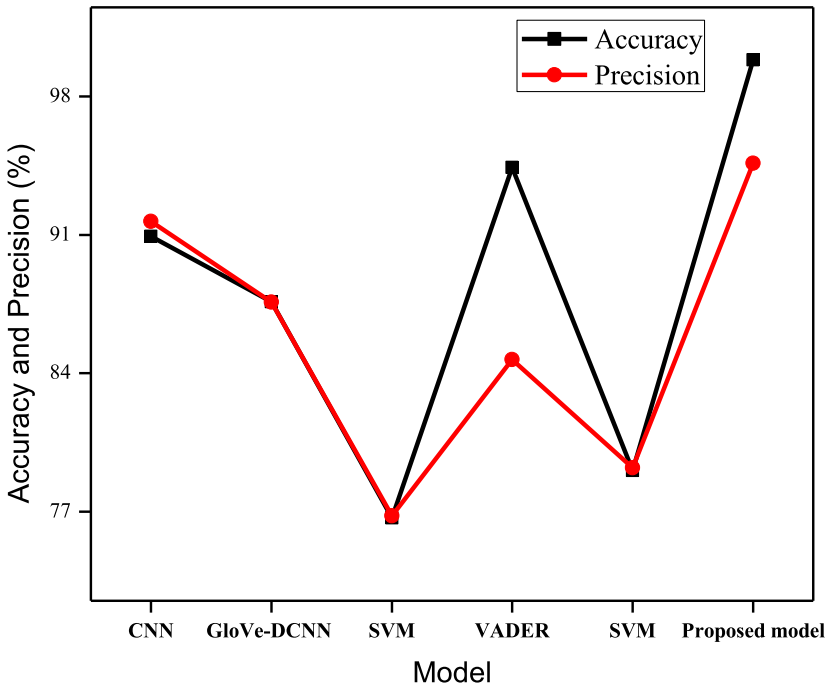


Fig. 8. Accuracy and precision of various models.

The authors have used deep convolution neural networks for sentiment classification on Twitter tweets. The proposed technique has obtained good performance over the existing techniques and the baseline model [52].

For the purpose of classifying the sentiment of Twitter tweets, [52] used deep convolution neural networks. In comparison to the baseline model and the already employed methodologies, the proposed technique has demonstrated good performance by training data from a substantial software infrastructure that used Deep Learning techniques like stacking and convolution for hierarchical representation. Table 11 compares the existing work with the suggested work, and Figure 7 depicts the performance analysis of the proposed model in terms of accuracy.

The comparison between the existing work and the proposed work is shown in Figure 8. Figure 8 shows that the accuracy and precision of the proposed model is higher compared to other proposed models [53]. The accuracy obtained through SVM is substantially lower; Although the VADER

model has excellent accuracy, its precision is lower to that of CNN, GloVe-DCNN, and the suggested model.

The prime discussions on the comparison between positive, negative, and neutral datasets are given here. On the positive dataset, the proposed model achieves a minimal error value of 0.017. The negative dataset has a minimum execution time of 0.33, while the neutral dataset has a minimum execution time of 0.06 [54]. On the positive dataset, the model achieves a minimal execution time of 0.44 seconds. On a malicious dataset, the minimal execution time is 0.37 seconds. On the malicious dataset, the greatest accuracy value is 67.14, whereas on the neutral dataset, it is 64.28. On the positive dataset, the model achieves the greatest recall value of 66.66. On the positive dataset, the model has the greatest F-measure value of 66.18. On the positive dataset, the suggested model achieves the best accuracy of 99.98 [55].

5 CONCLUSION

In the procedure, CNN's concept is used as a classifier to train the proposed system that helps in the sentiment analysis classification phase. According to the class, the system's training depends on the relevancy of the feature sets, and appropriate feature sets would train CNN. Therefore, in the classification section, stock market data can be easily classified with a trained structure regarding positive, negative, and neutral polarity. The outcome of the intended work on financial instruments in the market utilizing CNN has benefited greatly from a comparison of positive, negative, and neutral datasets on several parameters. When the present model is compared to the suggested model, it is clear that the latter produces far superior outcomes. The main inspiration in this work is that we are combining ABC and CNN is to accelerate model training and the use of the convolution function in the CNN model an optimistic forecast is created. The CNN model is used to identify the variables that will determine whether the stock price has risen or declined in the future. The designed technique predicts negative, positive, and neutral sentiment using stock news data taken from the Stock twists database. Therefore, findings in the study has improved the research, which is critical for stock market index prediction and have led to a better knowledge of the stock market's behavior in response to numerous happenings. The study's scope was limited owing to the scarcity of financial market statistics; as a result, it focused on a single stock market; nonetheless, the suggested model would be useful for other financial markets for a long time. Further studies in this topic might take into account country-specific causes such as economic and political issues, as well as time-specific determinants such as policy changes and implementation timelines.

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